Sonda Fourati

AI/ML Research Engineer Candidate & Research Master's Student in Information Technology Engineering at École de technologie supérieure (ÉTS Montréal)

+1 514 497 6116 sonda.fourati.1@ens.etsmtl.ca Sonda Fourati SondaFourati

Professional Summary

AI/ML researcher and Computer Systems Engineer with hands-on graduate research experience building, evaluating, and debugging machine-learning systems for computer vision, multimodal sensing, autonomous systems, and real-time AI pipelines. Experienced in Python, C/C++, TensorFlow/Keras, PyTorch-based vision workflows, Linux, experimental design, quantitative model/system evaluation, and research-to-prototype development. Published research in foundation models and multimodal autonomous driving, with a strong interest in developing reliable, high-performance AI systems and translating research ideas into production-quality implementations.

Education

École de Technologie Supérieure (ÉTS Montréal) **2025 – 2026**
Master's Degree in Information Technology *Université du Québec*

- GPA: 3.98/4.3

Mediterranean Institute of Technology — Medtech (ABET Accredited) **2022 – 2025**
Bachelor's Degree in Computer Systems Engineering *South Mediterranean University*

Higher Institute of Computer Science and Multimedia of Sfax (ISIMS) **2020 – 2022**
Integrated Preparatory Cycle *University of Sfax*

High School Diploma in Experimental Sciences **2020**
High School Diploma *El Hay High School*

Awards and Distinctions

Master's Research Scholarship **2025 – 2027**

- Research support scholarship during the master's program

Tuition Fee Exemption for International Students (ÉTS) **2025 – 2027**

- Exemption from additional tuition fees applied to international students

Mitacs GRA Scholarship **May 2024 – Aug 2024**

- Project topic: Improving autonomous driving using MLLMs

Best Student Award – ISS Project **May 2024**

- Project topic: PCB design

Best Student Award – ISS Project **May 2023**

- Project topic: Photographic Robot

International Ideathon Winner – Zayed University **March 2022**

- Competition theme: Reimagining the use of technology to enhance learning
- Links: Zayed University Award Video / Zayed University Award Photo

Current Project

Autonomous UGV Digital Twin | Master's Research Project, ÉTS Montréal **2025 – Present**

- Built an end-to-end Python/ROS 2 AI pipeline integrating image preprocessing, computer vision, deep-learning object detection, decision logic, real-time control, logging, and synchronized physical/simulated execution.
- Implemented and evaluated vision models for lane understanding and object detection on RGB streams, combining classical image processing with YOLO-based deep learning for robust autonomous decision-making.
- Designed controlled experiments and quantitative evaluation using latency, trajectory divergence, heading error, line-loss rate, success rate, and safety metrics to validate hypotheses and identify model/system failure modes.
- Developed reproducible sim-to-real and real-to-sim workflows in Gazebo, debugging multimodal pipelines across camera, LiDAR, odometry, ROS 2 QoS, networking, and NVIDIA Jetson edge inference.
- Analyzed performance bottlenecks and reliability issues in real-time inference/control loops, improving synchronization between real and simulated agents and supporting multi-robot experimental validation.

Professional Experience

Final Year Internship **May 2024 – Aug 2024**
ÉTS – LACIME Laboratory *Montreal, Canada*

- Conducted applied AI research on multimodal foundation-model-based autonomous driving, focusing on perception, reasoning, and decision-making under changing environmental conditions.
- Designed comparative experiments for an MLLM driving agent using CARLA and LimSim++ with camera and LiDAR inputs; analyzed behavior across weather conditions and investigated model failure cases.
- Contributed to the peer-reviewed survey "Foundation Models in Autonomous Driving: A Comprehensive Survey" link, synthesizing state-of-the-art foundation-model architectures, evaluation methods, and research challenges.
- Co-authored "A Novel MLLM-based Approach for Autonomous Driving in Different Weather Conditions" link, including experimental methodology, performance analysis, and scientific reporting.
- **Keywords:** Multimodal AI, Foundation Models, GPT-4o, Autonomous Driving, CARLA, LimSim++, Computer Vision

IoT Intern **July 2023 – Aug 2023**
Habemus! *Technopark, Sfax, Tunisia*

- Developed and deployed a remote air-quality monitoring pipeline using LoRaWAN and heterogeneous sensors, integrating embedded data acquisition, wireless transmission, and end-to-end system validation.
- **Keywords:** LoRaWAN, Arduino, Air Quality Monitoring

Software and AI Intern **May 2023 – July 2023**
Medtech in collaboration with Benten Company *Tunis, Tunisia*

- Implemented a conversational AI dialogue-management system using RASA, designing contextual conversation flows and integrating persistent user/session data with MongoDB.
- **Keywords:** RASA Framework, MongoDB

Software and AI Intern **July 2022 – Aug 2022**
LRIMA, Collège de Maisonneuve *Montreal, Canada*

- Improved the Alivecode platform through front-end engineering, debugging, and iterative interface enhancements to improve usability and reliability.
- Produced a promotional video for the ALIVEcode platform using Adobe Premiere Pro.
- **Keywords:** CSS, HTML, TypeScript, Adobe Premiere Pro

Selected Publications

Foundation Models in Autonomous Driving: A Comprehensive Survey **2026**

- Peer-reviewed survey covering multimodal foundation models, perception/reasoning architectures, datasets, evaluation strategies, and open challenges for autonomous driving.

A Novel MLLM-based Approach for Autonomous Driving in Different Weather Conditions **2025**

- Co-authored experimental study evaluating a multimodal large-language-model driving agent under varying environmental and sensing conditions.

Academic Projects

Image Captioning (Transport): LSTM/GRU/Transformer vs BLIP-2 **Sept. 2025 – Dec. 2025**

- Built an end-to-end image-captioning research pipeline on an MS COCO transport subset, including preprocessing, ResNet50 feature extraction, model training, inference, and reproducible evaluation.
- Implemented and compared LSTM, GRU, Transformer, and BLIP-2 approaches using BLEU, METEOR, ROUGE, CIDEr, and SPICE; analyzed quality/performance trade-offs and model errors.
- **Technologies:** Python, TensorFlow/Keras, ResNet50, Transformers, BLIP-2, NumPy, Pandas

PCB Design **Jan 2024 – May 2024**

- Designed a custom PCB tailored for an autonomous car system.
- Optimized layout, reduced signal interference, and improved system reliability and performance.
- **Technologies:** Altium Designer

Graph Theory: Path Planning **Apr 2023 – May 2023**

- Implemented and experimentally evaluated graph-based motion planning on a physical robot, testing path quality and execution behavior in real-world conditions.
- **Technologies:** Python, Pygame

Photographic Robot **Jan 2023 – May 2023**

- Designed and integrated a robotic platform supporting trajectory execution, image capture, and live video streaming, combining embedded control, sensing, and wireless communication.
- **Technologies:** Arduino, SolidWorks, ESP32, App Inventor

Part-Time Experience

Exam Invigilator **2025 – Present**

École de technologie supérieure (ÉTS) *Montréal, Canada*

- Supervising university examinations and ensuring compliance with institutional procedures and academic integrity policies
- Monitoring exam rooms, assisting students with procedural questions, and reporting irregularities when required
- Supporting the smooth and organized administration of examinations in a professional and impartial manner

Mitacs Globalink Mentor **April 2026 – Present**

Mitacs Globalink Program *Canada*

- Mentoring incoming international research interns in adapting to Canadian research environments, academic expectations, and lab workflows
- Providing technical guidance on research methodology, experiment design, coding practices, and AI/robotics tools used in research projects
- Supporting interns in project structuring, coding practices, and scientific communication to ensure successful research outcomes

Instructor **April 2026**

L'académie *Pointe-claire,Canada*

- Delivered a hands-on course on Microsoft Power Automate focused on workflow automation, process optimization, and real-world business applications
- Designed practical labs and exercises enabling students to build automated workflows for data handling, notifications, and system integration

Technical Skills

Programming: Python, C/C++, Java, JavaScript, SQL, HTML/CSS **Machine Learning / AI:** TensorFlow/Keras, PyTorch/Ultralytics, scikit-learn, Transformers, BLIP-2, YOLO, ResNet50, RASA **Computer Vision / Multimodal AI:** OpenCV, image preprocessing, object detection, multimodal camera/LiDAR pipelines, model evaluation, error analysis **Data / Experimentation:** NumPy, Pandas, controlled experiments, statistical analysis, performance metrics, reproducibility, research benchmarking **Systems / Robotics:** Linux, ROS 2, Gazebo, CARLA, NVIDIA Jetson, real-time systems, sensor integration, sim-to-real/real-to-sim workflows **Software Engineering:** Git/GitHub, debugging, modular development, Agile/Scrum, Docker (basic), Visual Studio Code **Databases / Embedded:** MySQL, PL/SQL, MongoDB, Arduino, STM32, Raspberry Pi, ESP32, LoRaWAN **Mathematics:** Statistics, Probability, Linear Algebra, Graph Theory, Signal Processing

Languages

Arabic: C2
French: C1
English: C1
German: A2

Mini Projects

- **Online Quiz Game:** Developed an online quiz game where a server sends questions and multiple clients compete to answer them while tracking scores and enforcing time limits.

Coursera Certificates

Ethics, Technology and Engineering **Jan 2024**
Eindhoven University of Technology

5G for Everyone **Jan 2024**
Qualcomm Academy

Excel Fundamentals Certificate **Nov 2023**
Corporate Finance Institute

Create Your First NoSQL Database with MongoDB and Compass **Apr 2023**
Coursera Project Network

Shortest Path Routing in Python **Mar 2023**
Coursera Project Network

Agile Development and Scrum **Mar 2023**
IBM Skills Network

Business Model Canvas for Entrepreneurs **Oct 2022**
Kennesaw State University

C Programming: Modular Programming and Memory Management **Oct 2022**
Dartmouth / Institut Mines-Télécom

Certificates

Coding with Python **Jan 2022**
GoMyCode

National Microsoft Technologies Camp **Mar 2021**
Microsoft ISIMS Club

National Status of Entrepreneur Student **Jun 2020**
Sfax University

Volunteer Experience

AIESEC Sfax **2020 – 2022**
Incoming Global Volunteer (IGV) and Outgoing Global Volunteer (OGV) member